

```
1: #include<stdio.h>
2: #include<stdlib.h>
3: #include<time.h>
4:
5: int main(void)
6: {
7:     int lotto[6] = {0};
8:     int i, j, k;
9:     srand((unsigned)time(NULL));
10:    printf("Before Sorting : ");
11:    for(i=0; i<6; i++)
12:    {
13:        do
14:        {
15:            lotto[i] = (rand() % 45 + 1);
16:            k = 0;
17:            for(j=0; j<i; j++)
18:            {
19:                if(lotto[i] == lotto[j])
20:                    k=1;
21:            }
22:        } while(k == 1);
23:        printf(" %d ", lotto[i]);
24:    }
25:
26:    printf("Lotto Numbers: ");
27:    for(i=0; i<5; i++)
28:    {
29:        for(j=i+1; j<6; j++)
30:        {
31:            if(lotto[i]>lotto[j])
32:            {
33:                int temp = lotto[i];
34:                lotto[i] = lotto[j];
35:                lotto[j] = temp;
36:            }
37:        }
38:    }
39:    for(i=0; i<6; i++)
40:    {
41:        printf(" %d ", lotto[i]);
42:    }
43:
44:    return 0;
45: }
```

```
1: #include<stdio.h>
2: #include<stdlib.h>
3: #include<time.h>
4: int main(void)
5: {
6:     srand((unsigned)time(NULL));
7:     int random[20] = {0};
8:     int i, j, k, l;
9:
10:    printf("---i\203\235i\204±ë\220\234 ë\202\234i\210\230 20ê°\234---\n");
11:    for(i=0; i<20; i++)
12:    {
13:        random[i] = (rand() % 9 + 1);
14:        printf(" %d ", random[i]);
15:    }
16:    printf("\n\n");
17:    printf("---i\210«i\236\220 i¶\234i\230\204
ë¹\210ë\217\204(Histogram)---\n");
18:
19:    for(l=0; l<=9; l++)
20:    {
21:        k = 0;
22:        for(i=0; i<20; i++)
23:        {
24:            if(random[i] == l)
25:            {
26:                k = k+1;
27:            }
28:        }
29:        printf("%d : %d\n", l, k);
30:    }
31:    return 0;
32: }
```

```
1: #include<stdio.h>
2: int isLeapYear(int year);
3: int getMonthDays(int year, int month);
4:
5: int main(void)
6: {
7:     unsigned int date=1, year, month, day, leap;
8:     char dayNames[7][3] = {{ 'S', 'u', 'n' }, { 'M', 'o', 'n' }, { 'T', 'u', 'e' }, { 'W',
'e', 'd' }, { 'T', 'h', 'u' }, { 'F', 'r', 'i' }, { 'S', 'a', 't' }};
9:     int calendar[6][7];
10:    printf("i\227°è\217\204i\231\200 i\233\224i\235\204
i\236\205ë ¥í\225\230i\204,i\232\224: ");
11:    scanf("%d %d", &year, &month);
12:    int totaldays=0, startmonth=0, startday;
13:    printf(" --- %d - %d --- \n", year, month);
14:    for(int i=0; i<7; i++)
15:    {
16:        for(int j=0; j<3; j++)
17:        {
18:            printf("%c", dayNames[i][j]);
19:        }
20:        printf(" ");
21:    }
22:    printf("\n");
23:    day = getMonthDays(year, month);
24:    leap = isLeapYear(year);
25:
26:
27:    for(int startyear=2000; startyear<year; startyear++)
28:    {
29:        if(isLeapYear(startyear) == 1)
30:            totaldays = totaldays + 366;
31:        else
32:            totaldays = totaldays + 365;
33:    }
34:
35:    for(startmonth=1; startmonth<month; startmonth++)
36:    {
37:        totaldays = totaldays + getMonthDays(year, startmonth);
38:    }
39:
40:    startday = (totaldays + 6)%7;
41:
42:    for(int i=0; i<6; i++)
43:    {
44:        for(int j=0; j<7; j++)
45:        {
46:            calendar[i][j] = 0;
47:        }
48:    }
49:
50:    date = 1;
51:    for(int i=0; i<6; i++)
52:    {
53:        for(int j=0; j<7; j++)
54:        {
55:            if(i==0 && j<startday)
56:                continue;
57:            if(date<=day)
58:            {
59:                calendar[i][j] = date;
60:                date++;
61:            }
62:        }
63:    }
64:
65:    for(int i=0; i<6; i++)
```

```
66:    {
67:        for(int j=0; j<7; j++)
68:        {
69:            if(calendar[i][j]==0)
70:                printf(" ");
71:            else
72:                printf("%3d", calendar[i][j]);
73:        }
74:        printf("\n");
75:    }
76:
77:    return 0;
78: }
79:
80: int isLeapYear(int year)
81: {
82:     if(year % 4 == 0 && year % 100 != 0)
83:         return 1;
84:     else if(year % 400 == 0)
85:         return 1;
86:     else
87:         return 0;
88: }
89:
90: int getMonthDays(int year, int month)
91: {
92:     int day = 0;
93:     if(month==1 || month==3 || month==5 || month==7 || month==8 || month==10
|| month==12)
94:         day = 31;
95:     else if(month==4 || month==6 || month==9 || month==11)
96:         day = 30;
97:     else if(month==2)
98:         if(isLeapYear(year)==1)
99:             day = 29;
100:        else
101:            day = 28;
102:    return day;
103: }
104:
105:
```

```
1: #include<stdio.h>
2: #include<stdlib.h>
3: #include<time.h>
4: int array[5][5];
5: void placemines(void);
6: void calculatemines(void);
7: void printboard(void);
8:
9: int main(void)
10: {
11:     printf("<<< Mine Sweeper Board >>>\n");
12:     placemines();
13:     calculatemines();
14:     printboard();
15:
16:     return 0;
17: }
18:
19: void placemines(void)
20: {
21:     srand((unsigned)time(NULL));
22:     for(int i=0; i<5; i++)
23:     {
24:         for(int j=0; j<5; j++)
25:         {
26:             array[i][j] = 0;
27:         }
28:     }
29:
30:
31:     for(int count=0; count<5; count++)
32:     {
33:         int i = rand() % 5;
34:         int j = rand() % 5;
35:
36:         if(array[i][j] == -1)
37:         {
38:             count--;
39:         }
40:         else
41:             array[i][j] = -1;
42:     }
43: }
44:
45: void calculatemines(void)
46: {
47:     int count;
48:     for(int i=0; i<5; i++)
49:     {
50:         for(int j=0; j<5; j++)
51:         {
52:             if(array[i][j] == 0)
53:             {
54:                 int count = 0;
55:                 for(int ni=i-1; ni<i+2; ni++)
56:                 {
57:                     for(int nj=j-1; nj<j+2; nj++)
58:                     {
59:                         if(nj<0 || ni<0 || nj>4 || ni>4)
60:                             continue;
61:                         if(array[ni][nj] == -1)
62:                         {
63:                             count++;
64:                         }
65:                     }
66:                     array[i][j] = count;
67:                 }

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```
68:             }
69:         }
70:     }
71: }
72:
73: void printboard(void)
74: {
75:     for(int i=0; i<5; i++)
76:     {
77:         for(int j=0; j<5; j++)
78:         {
79:             printf("%3d", array[i][j]);
80:         }
81:         printf("\n");
82:     }
83: }
```

```
1: #include<stdio.h>
2: #include<stdlib.h>
3: #include<time.h>
4:
5: void initmatrix(char mat[10][10], int n);
6: void printmatrix(char mat[10][10], int n);
7: void rotate90(char mat[10][10], int n);
8: void rotate180(char mat[10][10], int n);
9: void rotate270(char mat[10][10], int n);
10: char mat[10][10];
11: char matresult[10][10];
12:
13: int main(void)
14: {
15:     int n;
16:     printf("í\226\211ë ~ì\235\230 í\201~ê,° Nì\235\204
i\236\205ë ¥í\225\230ì\204,i\232\224 (1~10): ");
17:     scanf("%d", &n);
18:     printf("[Original Matrix]\n");
19:     initmatrix(mat, n);
20:     printf("[90 Degree Rotation]\n");
21:     rotate90(mat, n);
22:     printf("[180 Degree Rotation]\n");
23:     rotate180(mat,n);
24:     printf("[270 Degree Rotation]\n");
25:     rotate270(mat, n);
26:     return 0;
27: }
28:
29:
30:
31: void initmatrix(char mat[10][10], int n)
32: {
33:
34:     srand((unsigned)time(NULL));
35:     for(int i=0; i<n; i++)
36:     {
37:         for(int j=0; j<n; j++)
38:         {
39:             mat[i][j] = 'A' + (rand() % 26);
40:             printf(" %c ", mat[i][j]);
41:         }
42:         printf("\n");
43:     }
44: }
45:
46: void rotate90(char mat[10][10], int n)
47: {
48:     int i, j;
49:     for(j=0; j<n; j++)
50:     {
51:         for(i=n-1; i>=0; i--)
52:         {
53:             printf(" %c ", mat[i][j]);
54:         }
55:         printf("\n");
56:     }
57: }
58: }
59:
60: void rotate180(char mat[10][10], int n)
61: {
62:     int i, j;
63:
64:     for(i=n-1; i>=0; i--)
65:     {
66:         for(j=n-1; j>=0; j--)
```

```
67:         {
68:             printf(" %c ", mat[i][j]);
69:         }
70:         printf("\n");
71:     }
72: }
73:
74: void rotate270(char mat[10][10], int n)
75: {
76:     int i, j;
77:
78:     for(j=n-1; j>=0; j--)
79:     {
80:         for(i=0; i<n; i++)
81:         {
82:             printf(" %c ", mat[i][j]);
83:         }
84:         printf("\n");
85:     }
86: }
87:
```